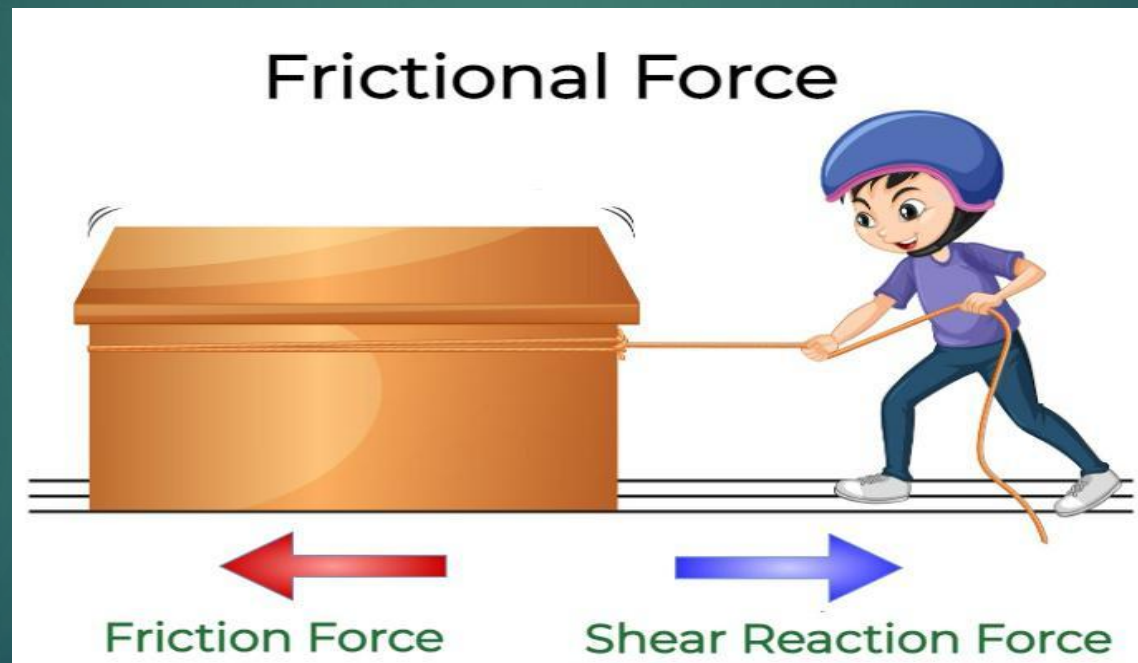


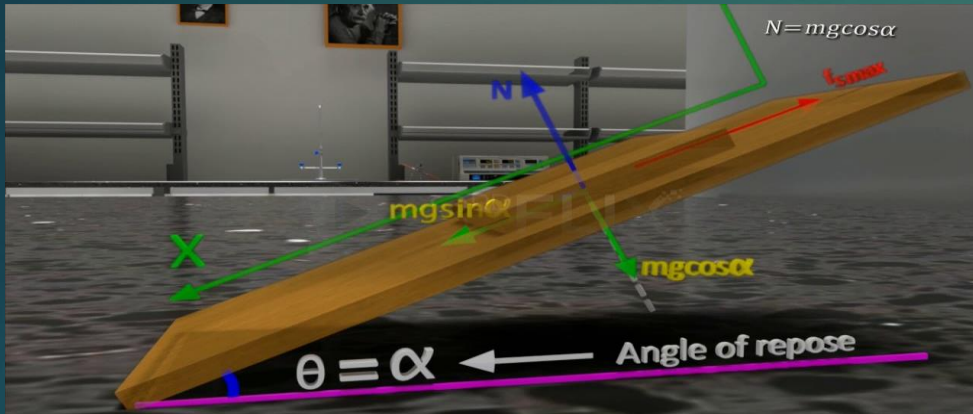
Experiment (5)

Friction

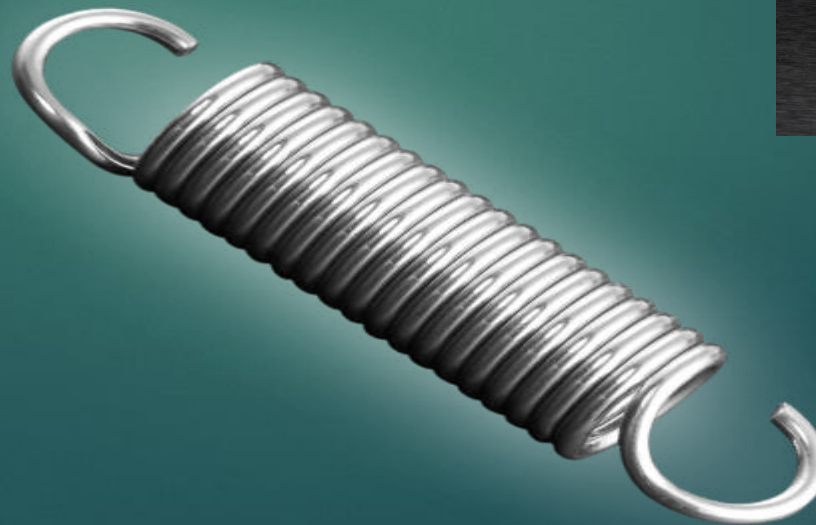


Apparatus

Incline Surface



Spring

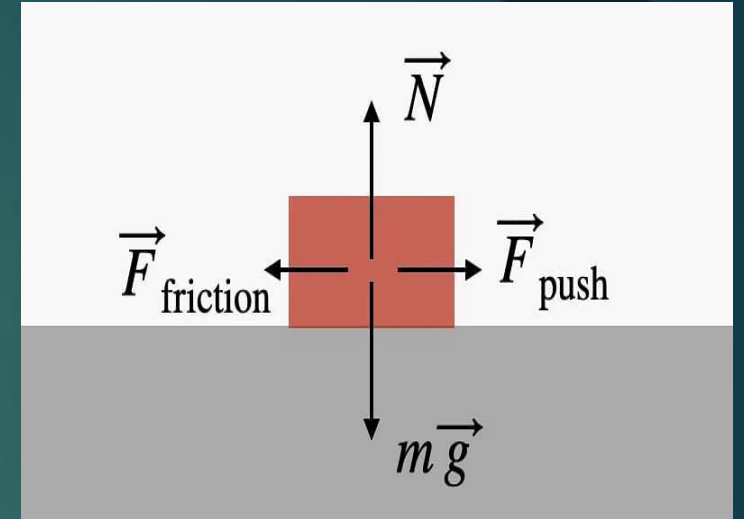


Weights



Introduction

- ▶ When two solid surfaces come in contact with each other there will be forces acting between the surfaces. These “contact forces” can be broken up into two components: a Normal Force which is perpendicular to the surface and prevents the surfaces from passing through each other and Friction which acts along the surface and prevents or hinders relative motion of the two surfaces



Theory

Part (1): Static Friction

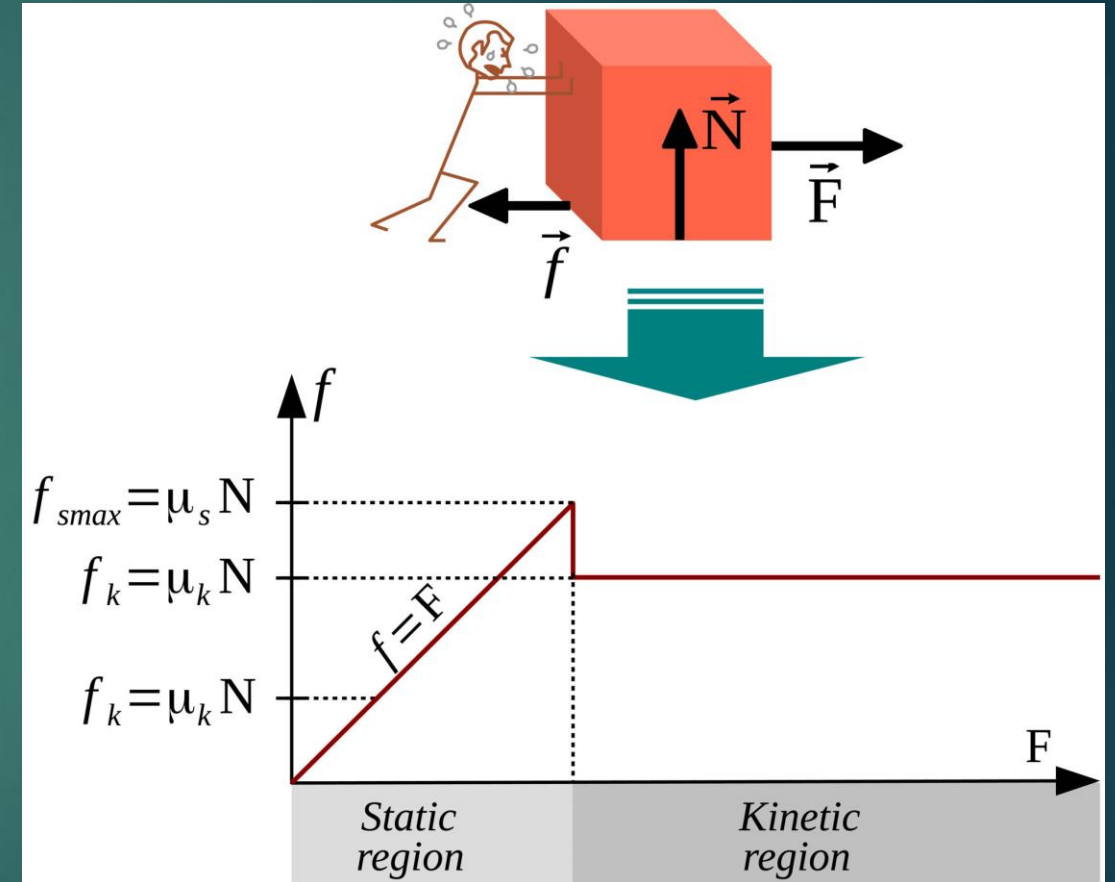
When we attempt to make two surfaces slide relative to each other, for example by trying to push a book across the table, the force of friction will oppose the relative motion between the two surfaces. This means that as we increase the force with which we push the book, the force of friction will increase until it has reached a limiting value. When the two surfaces are not (yet) moving relative to each other, we call the force of friction static friction. To describe static friction mathematically, we can model it with the equation

$$f_s \leq \mu_s N \dots \dots \dots (1)$$

Where f_s : The force of friction

μ_s : The coefficient of static friction and it is a number that depends on the materials and roughness of the two surfaces

N : The normal force between the two surfaces.



Consider an inclined plane, as shown in figure below, whose angle, θ , can be adjusted. If a block is placed on the plane while it is horizontal and the angle of the plane is slowly increased eventually the block will begin to slide. This is the point at which the limit of static friction has been reached. We can write the forces on the block in terms of components along (tangent to) the plane and components perpendicular (normal) to the plane: Perpendicular to the plane the acceleration is always zero and Newton's Laws tell us that

$$N - mg\cos\theta = ma_y \Rightarrow N = mg\cos\theta \dots\dots (2)$$

Before the block slides, friction is less than or equal to the static limit and there is no acceleration along the plane either. Thus the forces along the plane are given by:

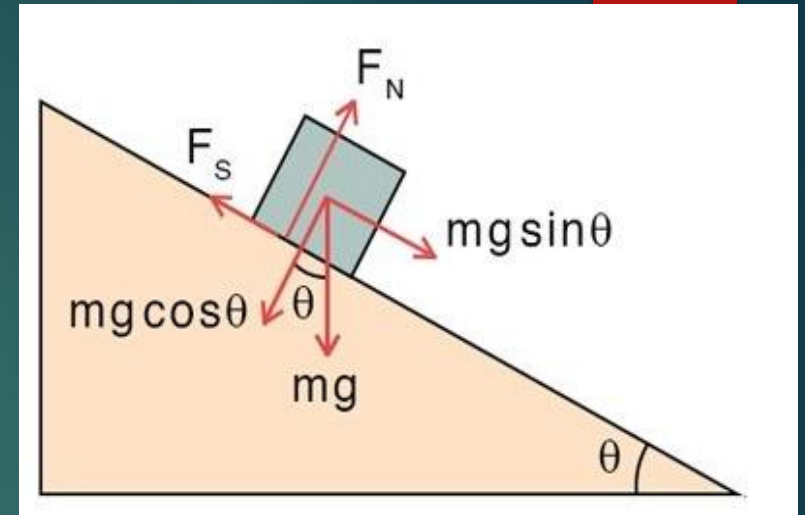
$$mg\sin\theta - F_s = ma_x \Rightarrow F_s = mg\sin\theta \dots\dots\dots (3)$$

In our model, $F_s \leq \mu_s N$ and therefore $F_s \leq \mu_s mg \cos\theta$

The force of friction increases as the angle increases until the maximum value is reached. Exactly at this angle, acceleration is still zero and:

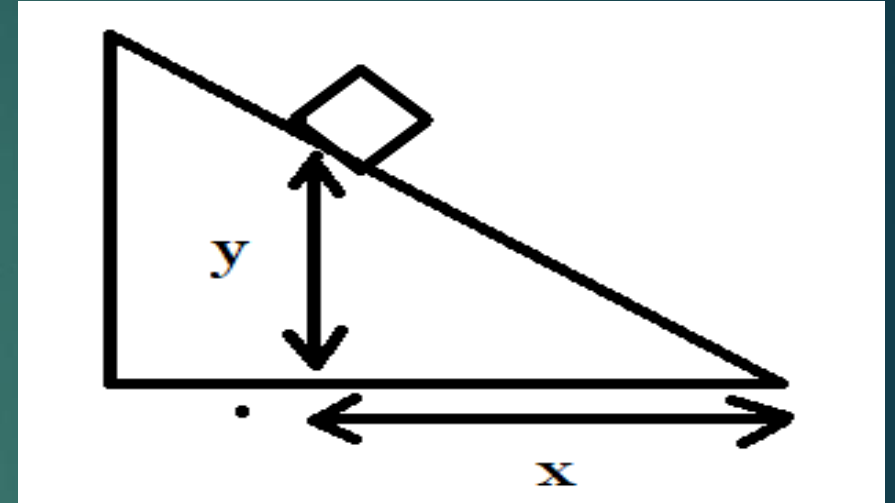
$$F_s = mg\sin\theta_s = \mu_s mg\cos\theta_s$$

$$\mu_s = \tan\theta_s \dots\dots\dots (4)$$



Procedure

1. Place the block with its large surface down on the board.
2. Adjust the angle of the board until the block just begins to slide on its own. Record the angle at which it slips as θ_s in the appropriate by measuring the height y and the distance x Data Table (1).
3. Repeat the procedure at least three more times.
4. Determine μ_s for the four measurements and then calculate the average value of μ_s determined by this method.



Part (2): Kinetic Friction

After the surfaces begin to slide relative to each other, friction holds a constant value and, while it can't prevent relative motion, it will hinder or slow down this motion. This type of friction is called kinetic friction and is represented by the equation:

$$f_k \leq \mu_k N \dots \dots \dots (5)$$

Where,

F_k : The force of friction

μ_k : The coefficient of kinetic friction and N is again the normal force.

If we attach a mass M to a spring of spring constant K and natural length L_0 , as shown in shown, and the spring is stretched a distance x_1 then released, it will come back to a distance x_2 . For such a system, due to the kinetic friction force, the mechanical energy will not be conserved as expressed in the following equation:

$$W_{fr} = E_f - E_i \dots \dots \dots (6)$$

W_{fr} : Work done by force of kinetic friction.

E_i : Initial mechanical energy.

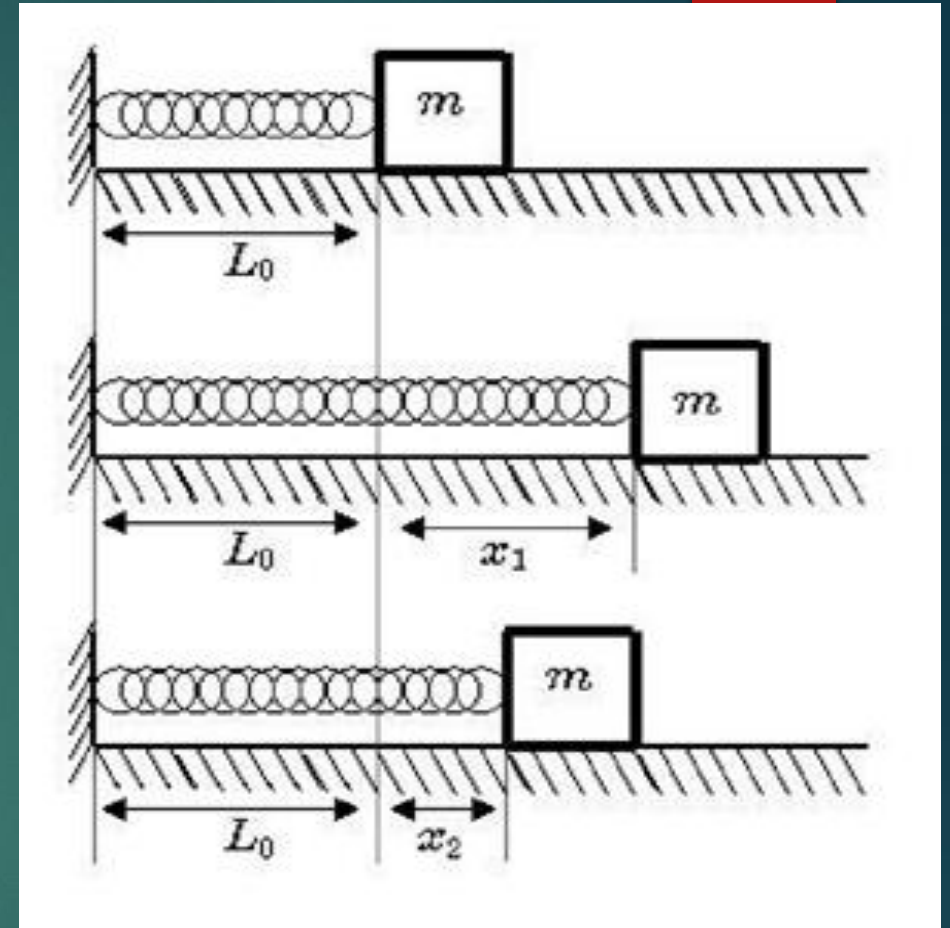
E_f : Final mechanical energy.

$$W_{fr} = \mu_k mg \Delta x, \quad E_i = \frac{1}{2} K x_1^2, \quad E_f = \frac{1}{2} K x_2^2$$

$$-\mu_k mg \Delta x = \frac{1}{2} K x_2^2 - \frac{1}{2} K x_1^2 \dots \dots \dots (7)$$

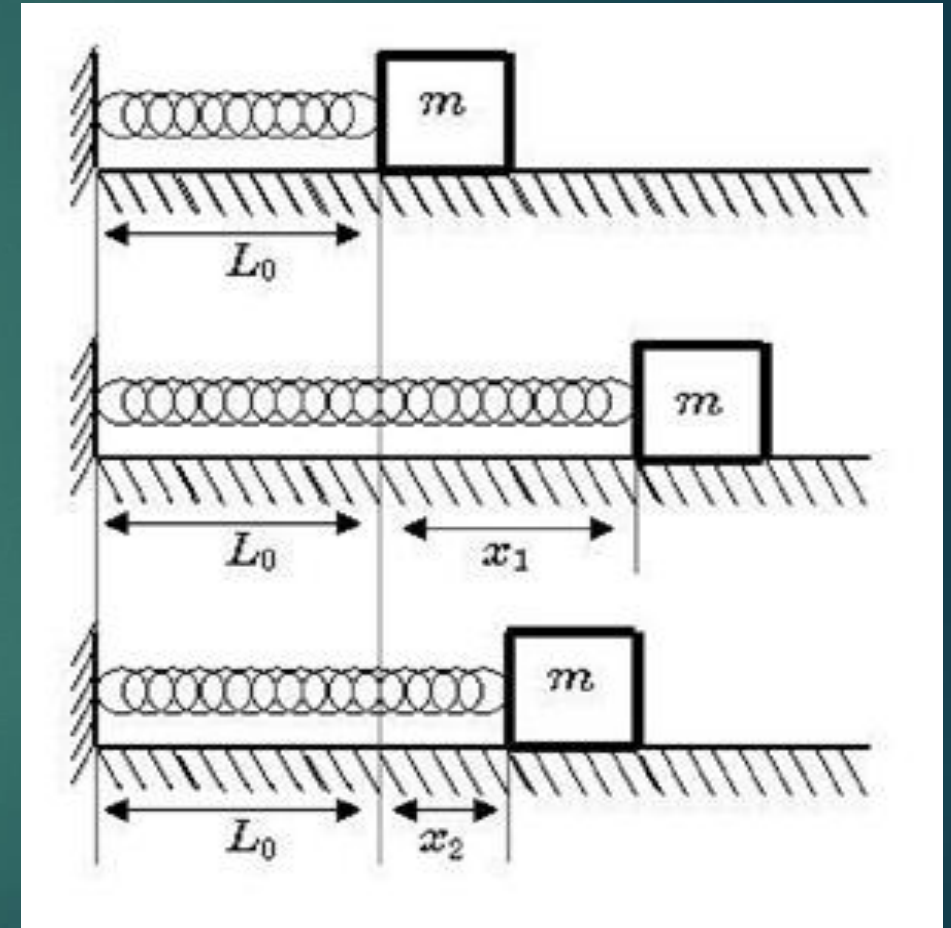
$$-\mu_k mg (x_1 - x_2) = \frac{1}{2} K (x_1 + x_2) (x_2 - x_1)$$

$$\mu_k mg = \frac{1}{2} K (x_1 + x_2) \dots \dots \dots (8)$$



Procedure

1. Assemble the apparatus as shown in Figure above
2. Measure the natural length of the spring L_0 and record it in the table
3. Add a masses on the wooden board and Pull the mass to the right to a new position and record the extension X_1 in the spring.
4. Release the spring from rest and record the final extension when coming to a stop X_2 .
5. Repeat steps 3 and 4 for four different masses and record X_1 and X_2 for each mass.
6. Record your measurements in a table
7. Plot on the y-axis the force (mg) and ($x_1 + x_2$) on the x-axis.
8. Obtain the best straight line that represents your experiment points.
9. Use the graph to find μ_k .
10. What is the experiment al error in μ_k ?



THANK YOU
